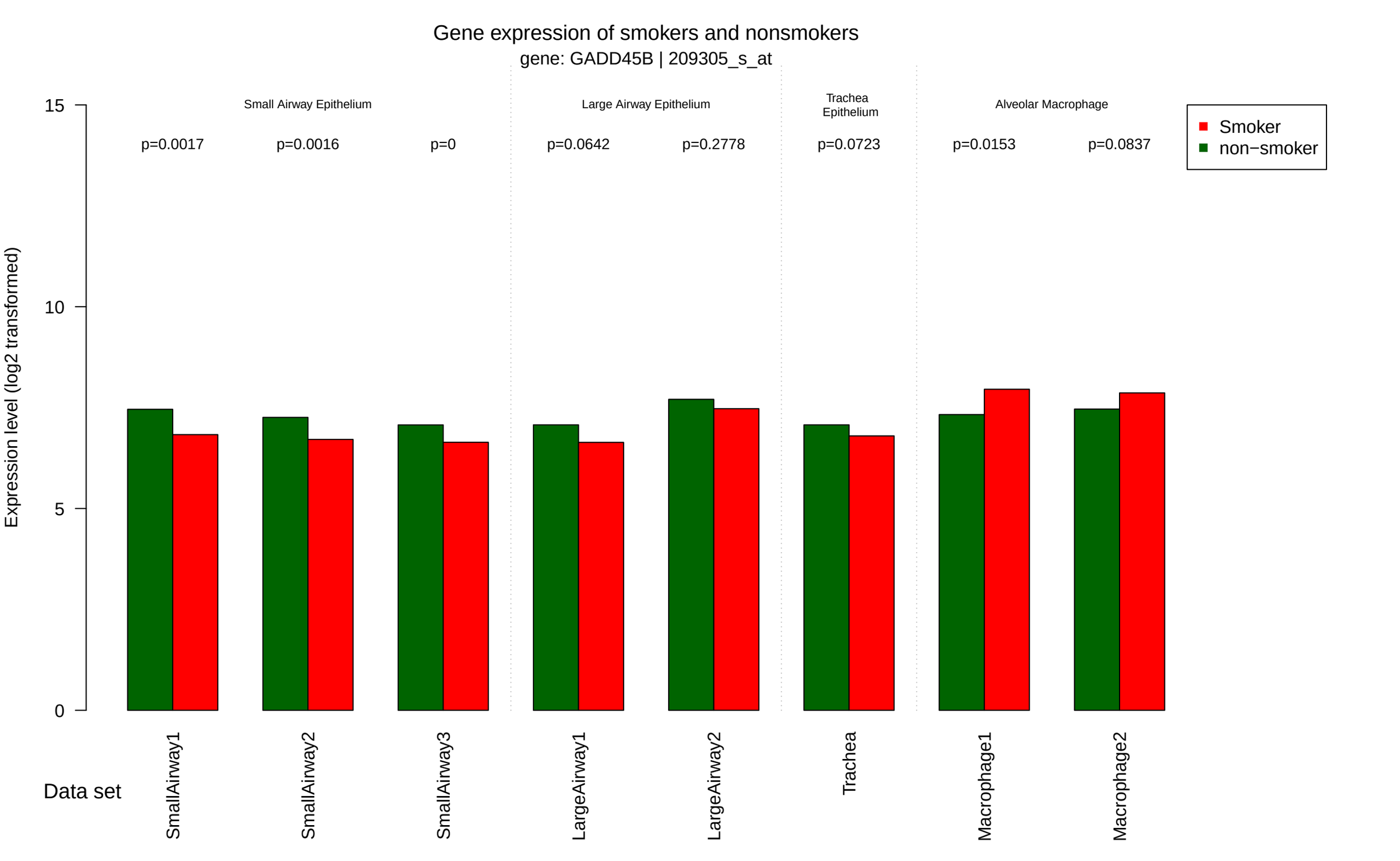
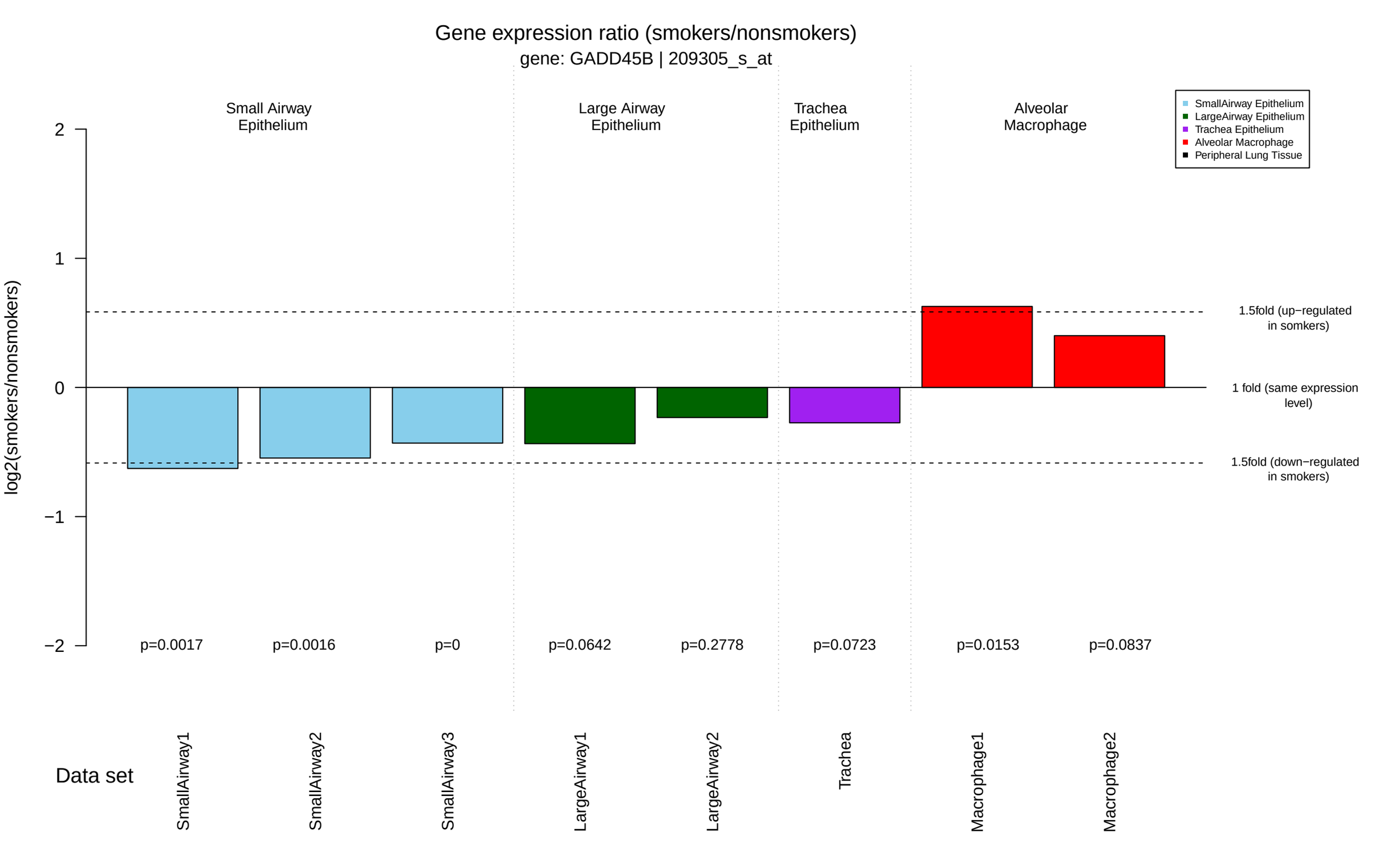
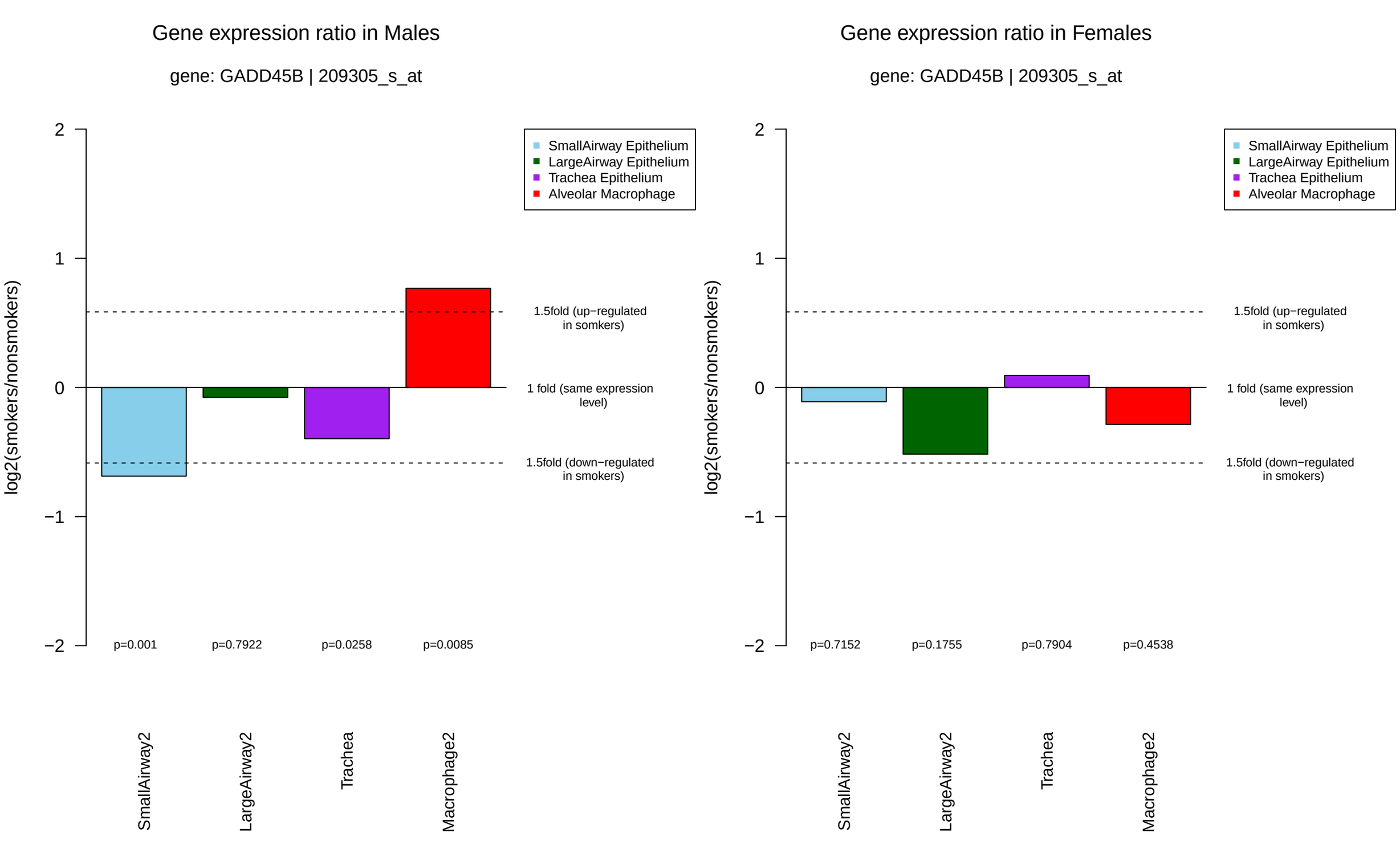
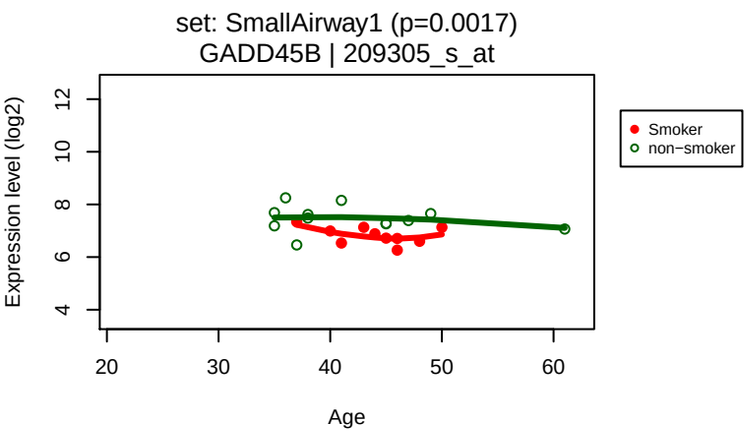
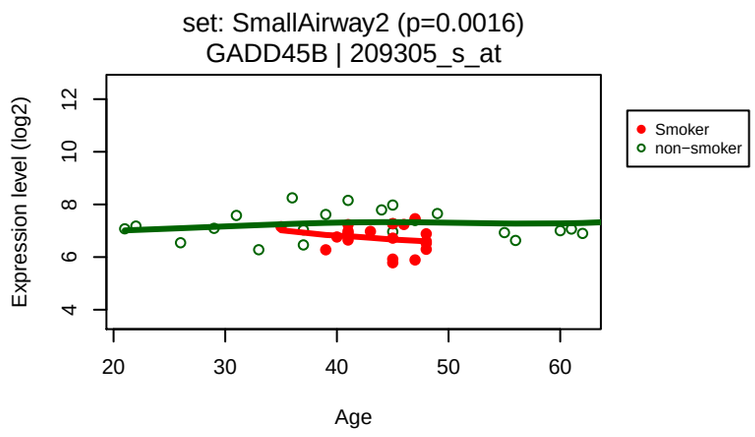
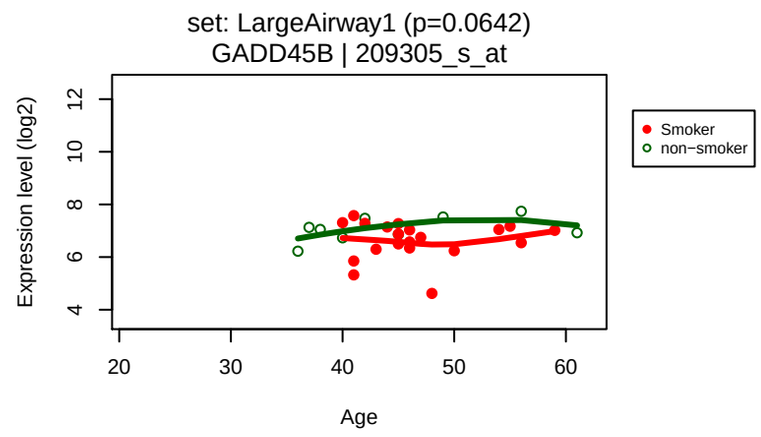

Gene expression of smokers and nonsmokers
gene: GADD45B | 209305_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0017
p=0.0016
p=0
p=0.0642
p=0.2778
p=0.0723
p=0.0153
p=0.0837
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

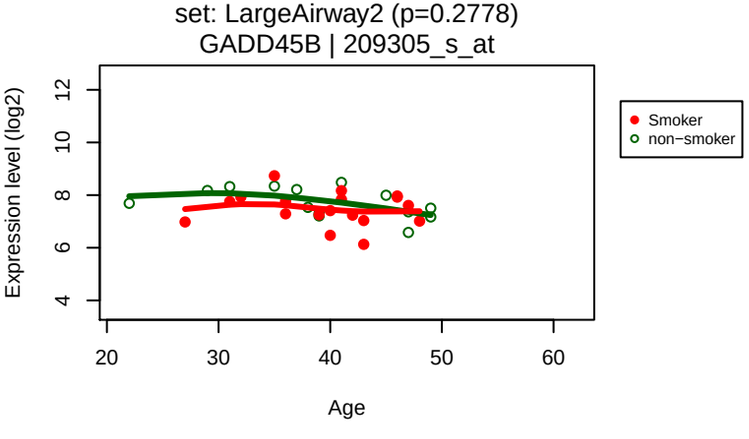

Gene expression ratio (smokers/nonsmokers)
gene: GADD45B | 209305_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0017
p=0.0016
p=0
p=0.0642
p=0.2778
p=0.0723
p=0.0153
p=0.0837
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

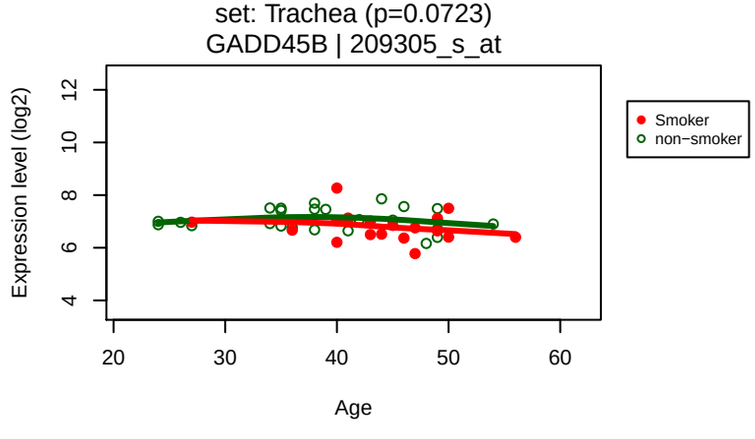

Gene expression ratio in Males
gene: GADD45B | 209305_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.001
p=0.7922
p=0.0258
p=0.0085
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GADD45B | 209305_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7152
p=0.1755
p=0.7904
p=0.4538
SmallAirway2
LargeAirway2
Trachea
Macrophage2

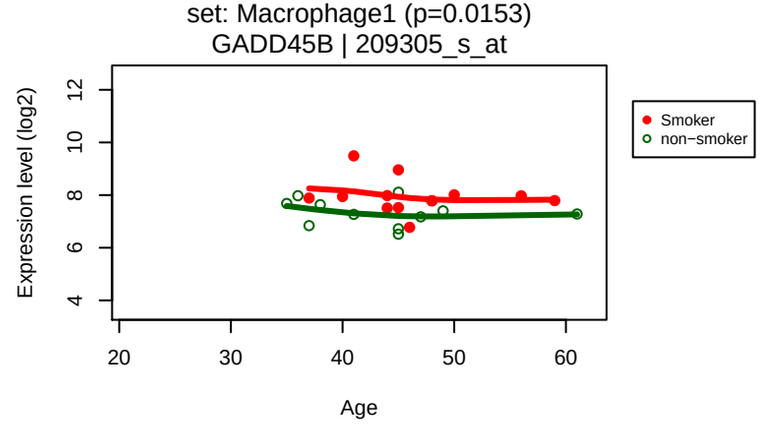

set: SmallAirway1 (p=0.0017)
GADD45B | 209305_s_at
Expression level (log2)
Age
Smoker
non−smoker

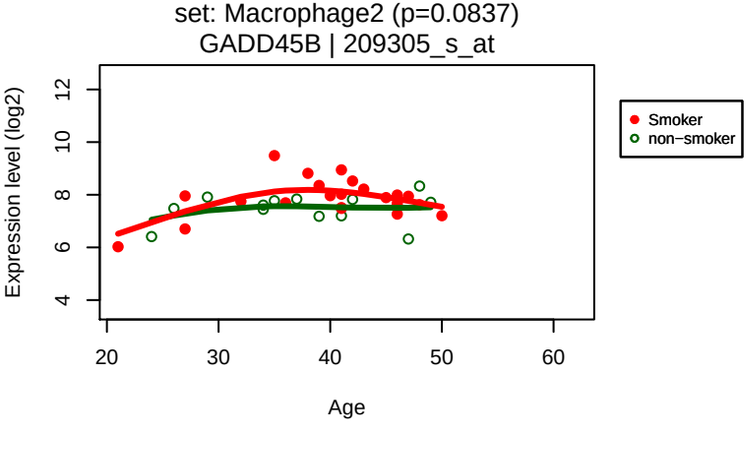

set: SmallAirway2 (p=0.0016)
GADD45B | 209305_s_at
Expression level (log2)
Age
Smoker
non−smoker

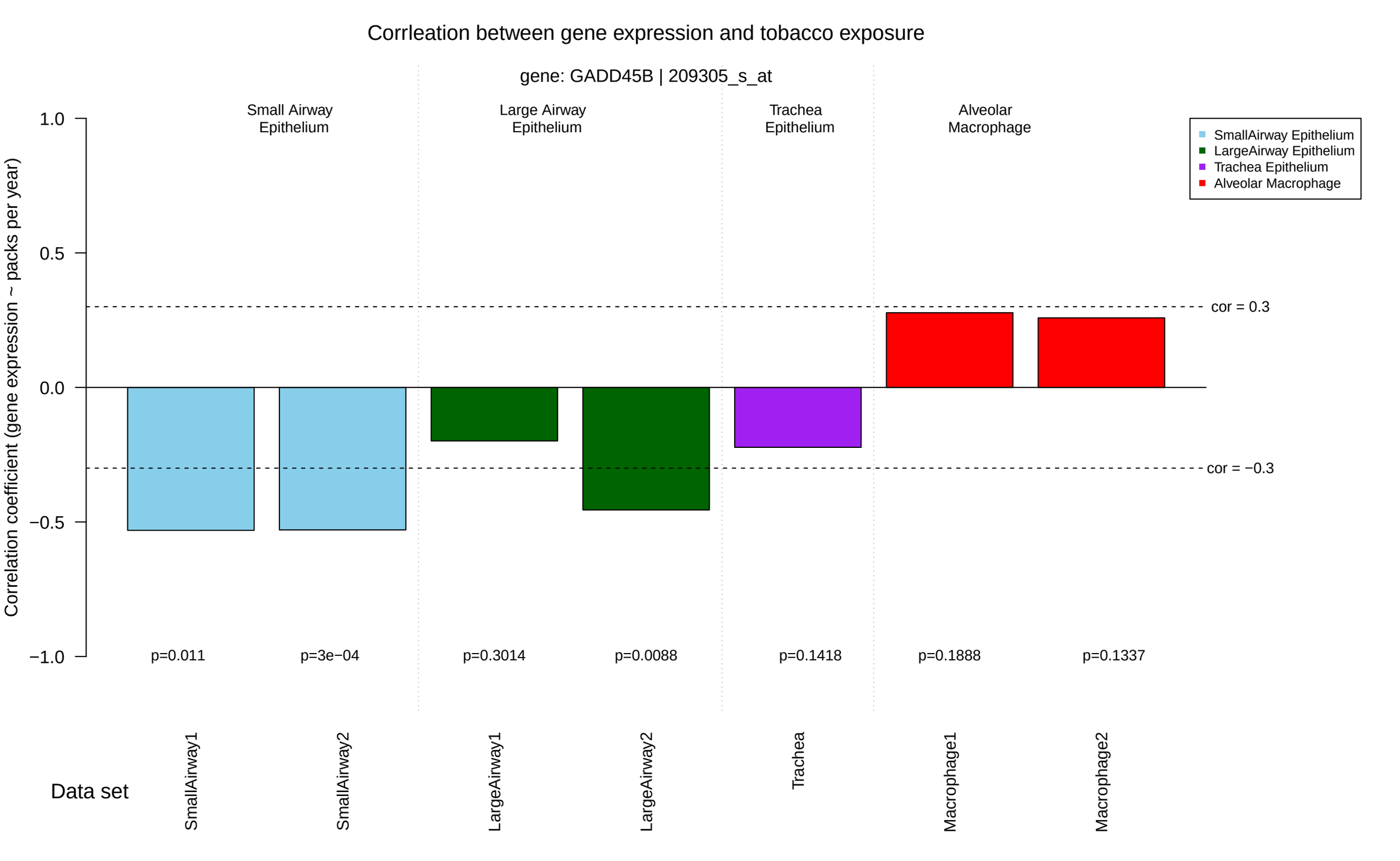

set: LargeAirway1 (p=0.0642)
GADD45B | 209305_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2778)
GADD45B | 209305_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0723)
GADD45B | 209305_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0153)
GADD45B | 209305_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0837)
GADD45B | 209305_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GADD45B | 209305_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.011
p=3e−04
p=0.3014
p=0.0088
p=0.1418
p=0.1888
p=0.1337
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2